



PRESENTATION

TBDA - Week 4



TEAM PRESENTATION - GROUP 3



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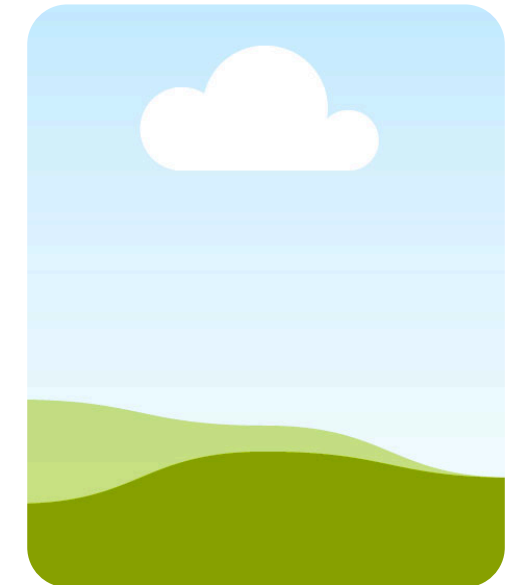
Marie Heyse



Riccardo Tardanico



Rita Piepoli



Sophie Scharpenberg



RESPONSIBILITIES

Sub-team	Team members	Core responsibility	Later shift (from week 6)
A. Coordination & Documentation	Marie Heyse	Project lead, weekly planning, experiment log (auditing), Moodle deliverables, weekly review slides	Final report & DataPack assembly
B. Data Engineering	Joos Van Lommel, Catherine Ma, Anthony Rivière	Ingest all Excel exports, unify schema & timestamps, resampling, detect/correct counter resets, load into PostgreSQL	1-2 members → feature engineering with Modelling
C. Domain & Semantic Modelling	María Herraiz, Riccardo Tardanico	Device & zone inventory (ID → floor → zone → subsystem), data dictionary, ontology (building/floor/zone/sensor), SPARQL queries aggregating energy per space	Support EDA with zone-level aggregations
D. Exploratory Analysis & Data Quality	Gaia, Ludovica	Consumption profiles (daily/weekly), gaps, outliers, links between energy and environmental data	Join Modelling (model diagnostics, error analysis)
E. Modelling & Forecasting	Mavii Cimaglia, Rita Piepoli	Define target & forecast horizon, baselines, candidate models, evaluation protocol	Core modelling team grows to 4-5 people
F. Visualisation & Communication	Kacper Vogel	Dashboard, visual identity, storyline "see the evidence"	Final presentation & dissemination material



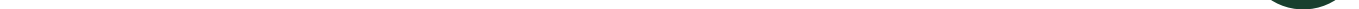


TASKS WEEK 3-4

- ▶ **A:** Set up the repo and shared drive. Create the experiment-log template (date, data used, model, parameters, metric, conclusion). Build a timeline to Nov 30.
 - ▶ **B:** Inventory all Excel files (which floor/zone, which devices, time span, sampling interval). Write a first loader script to one tidy format: timestamp | device_id | zone | metric | value.
 - ▶ **C:** Build the device-zone mapping table from the floor plans. Decode the telemetry key naming (phase/line/metric). Draft the ontology classes.
 - ▶ **D:** Run first plots for 2-3 devices. Quantify missing data, sampling irregularities and visible reset events.
 - ▶ **E:** Propose the prediction question. For example: hourly/daily energy per zone or per subsystem (lighting/HVAC/other), 24h-ahead. Also list candidate external data: weather, holidays, academic calendar.
 - ▶ **F:** Set up the presentation template and the first version of the architecture/pipeline diagram in the team's own words.
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TOOLS TO USE

Activity	Tools
Coordination	GitHub Projects (tasks), WhatsApp (communication), Google Drive/OneDrive (files and the sharing of sheets)
Version control & code	GitHub, Python in Jupyter/VS Code
Data ingestion & cleaning	Python: pandas (python library) and openpyxl (installation)
Storage	PostgreSQL
Semantic layer	Protégé (ontology design) → Brick Schema and Apache Jena Fuseki to demo live queries
EDA	matplotlib/seaborn/plotly → plottinglibraries , ydata-profiling for quick data-quality reports
External data	Open-Meteo → Madrid weather, Madrid/UPM holiday calendar
Modelling	scikit-learn, statsmodels (SARIMA), LightGBM/XGBoost; time-series cross-validation
Experiment tracking (auditing)	MLflow → automatically logs every model run
Methodology	CRISP-DM as the overall framework
Visualisation & communication	Power BI(dashboard) and Canva (presentations)



MAIN FINDINGS

- ▶ Exploration of the data
- ▶ We found that there's data missing for temperature, pressure, ... (indicated with 0)





TIMELINE



Week 5

Machine learning preparation

- lag matrix creation
- calculate rolling features
- warm up cut

Week 6

- scikit - learn pipeline creation

Week 7

- forecasted data manipulation



Q&A

